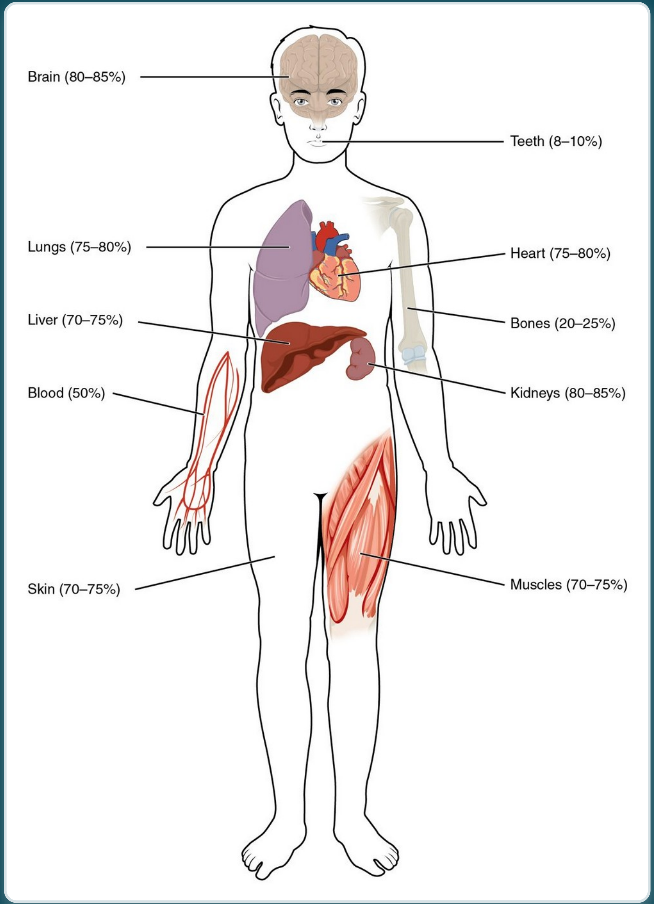
CHAPTER 26

Fluid, Electrolyte, and Acid-Base Balance

Keeping the body's water, ions, and pH in balance.



Anatomy & Physiology · Broward-Miami Health Institute

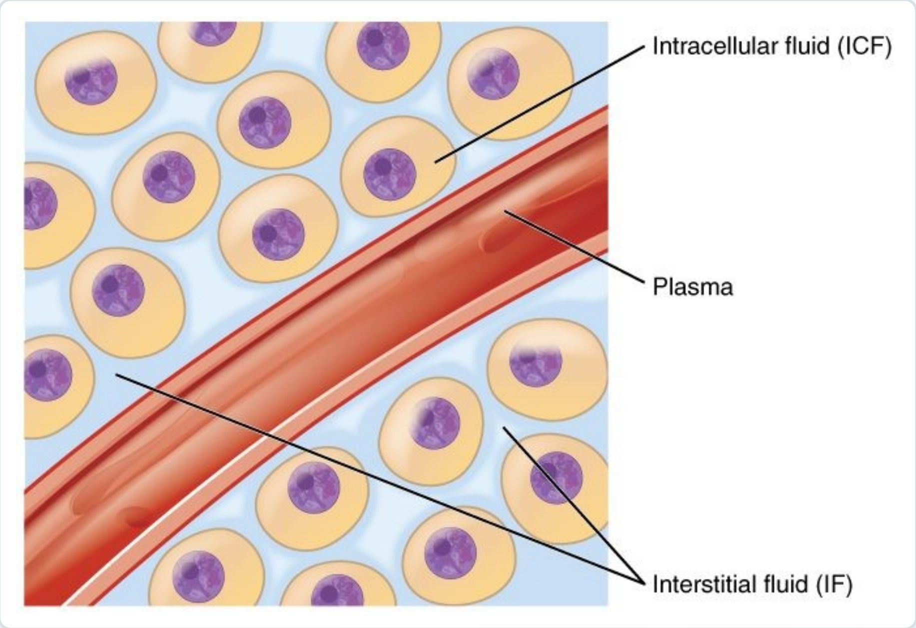
FACULTY EDITION

CHAPTER 26

Learning Objectives

After studying this chapter, students will be able to:

1. Identify the body’s main fluid compartments
2. Define plasma osmolality and identify two ways in which plasma osmolality is maintained
3. Identify the six ions most important to the function of the body
4. Define buffer and discuss the role of buffers in the body
5. Explain why bicarbonate must be conserved rather than reabsorbed in the kidney



The body's fluid compartments.

26.1

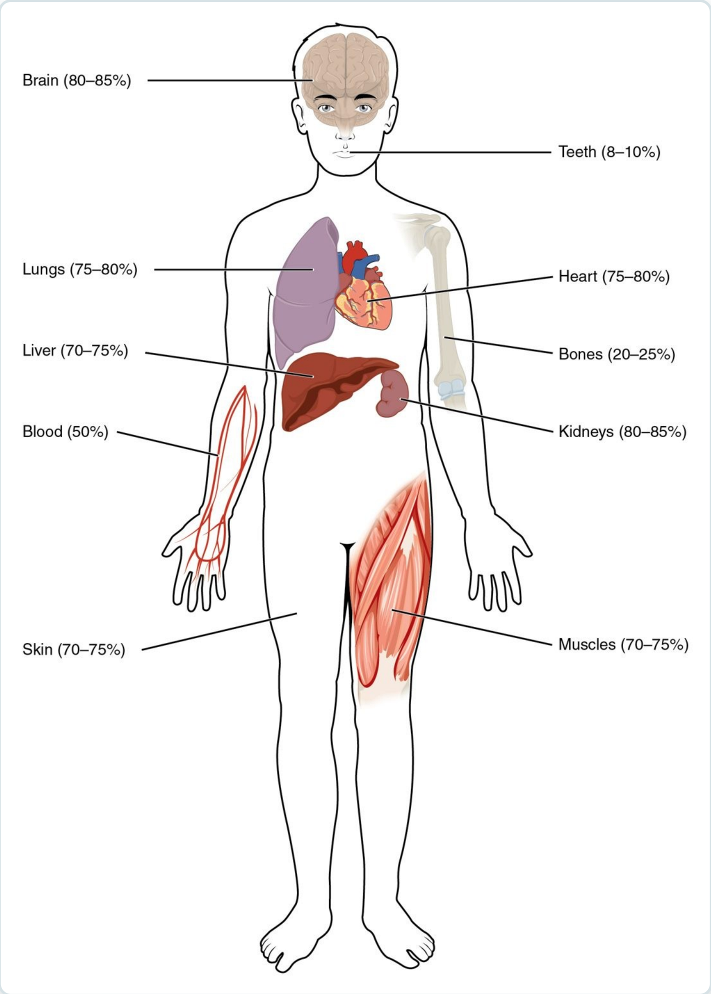
SECTION

Body Fluids and Fluid Compartments

Where the body's water is and how it is distributed.

SECTION 26.1

Body Water Content



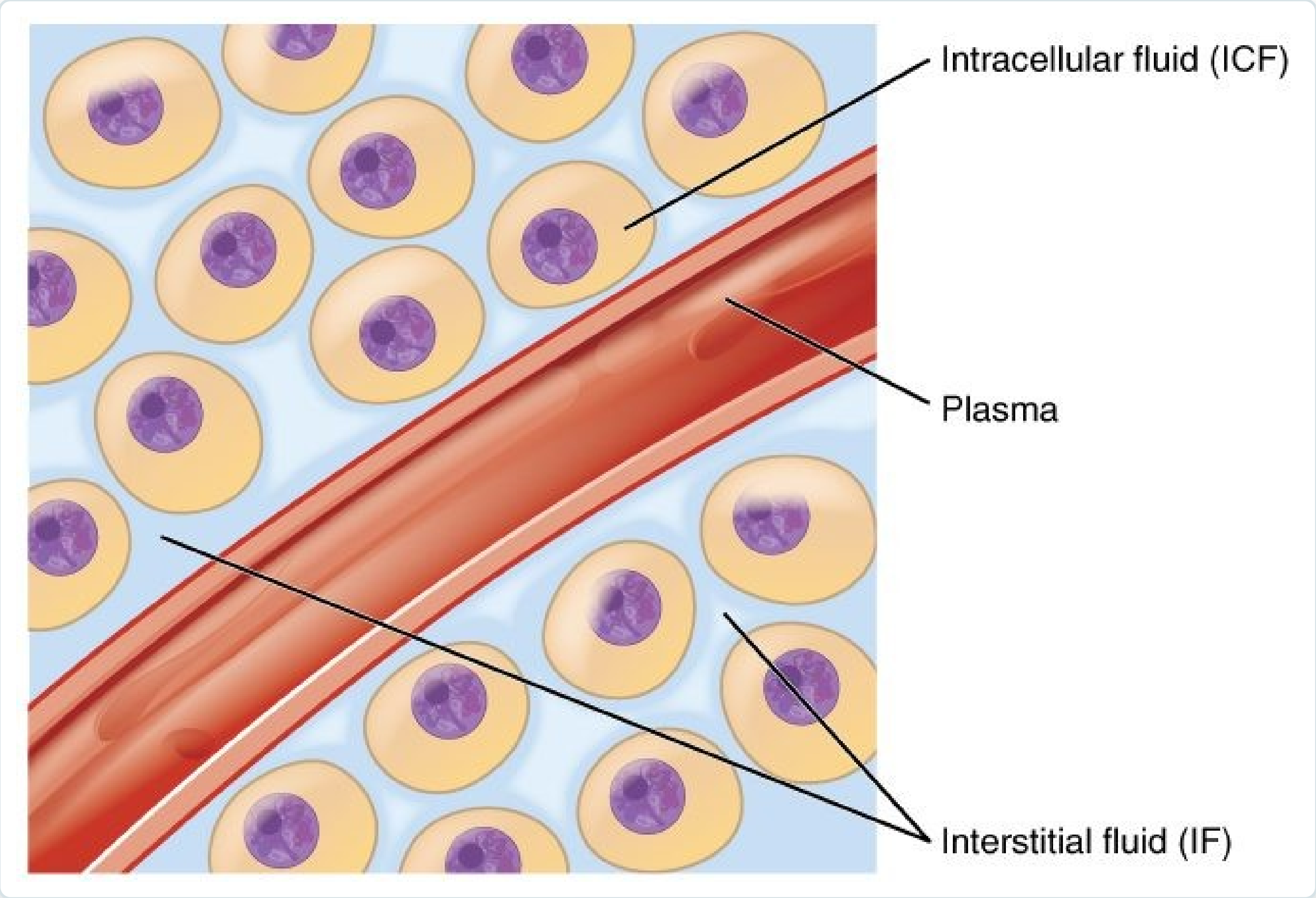
Body water content

- Water is most of the body
- About 55-60% of body weight
- Varies by tissue and age
- Highest in muscle, lowest in fat

SECTION 26.1

Fluid Compartments

- Body fluid sits in two compartments
- Intracellular fluid (inside cells)
- Extracellular fluid (outside cells)
- ECF = plasma + interstitial fluid



The body's fluid compartments

26.2

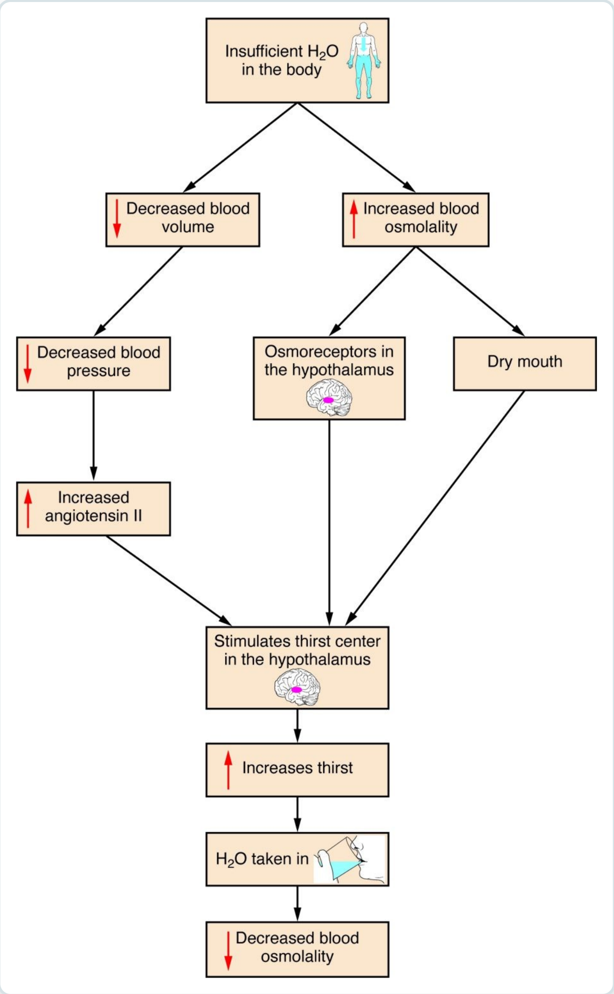
SECTION

Water Balance

How the body keeps water intake and loss in balance.

SECTION 26.2

Water Balance



Regulation of water balance

- Intake must match output
- Thirst drives intake
- ADH reduces water loss
- The kidneys are the main regulator

SECTION 26.2

Dehydration

- Fluid loss exceeds intake
- Common with vomiting, diarrhea, fever
- Signs: poor skin turgor, dark urine
- Dangerous if severe

Signs of dehydration

26.3

SECTION

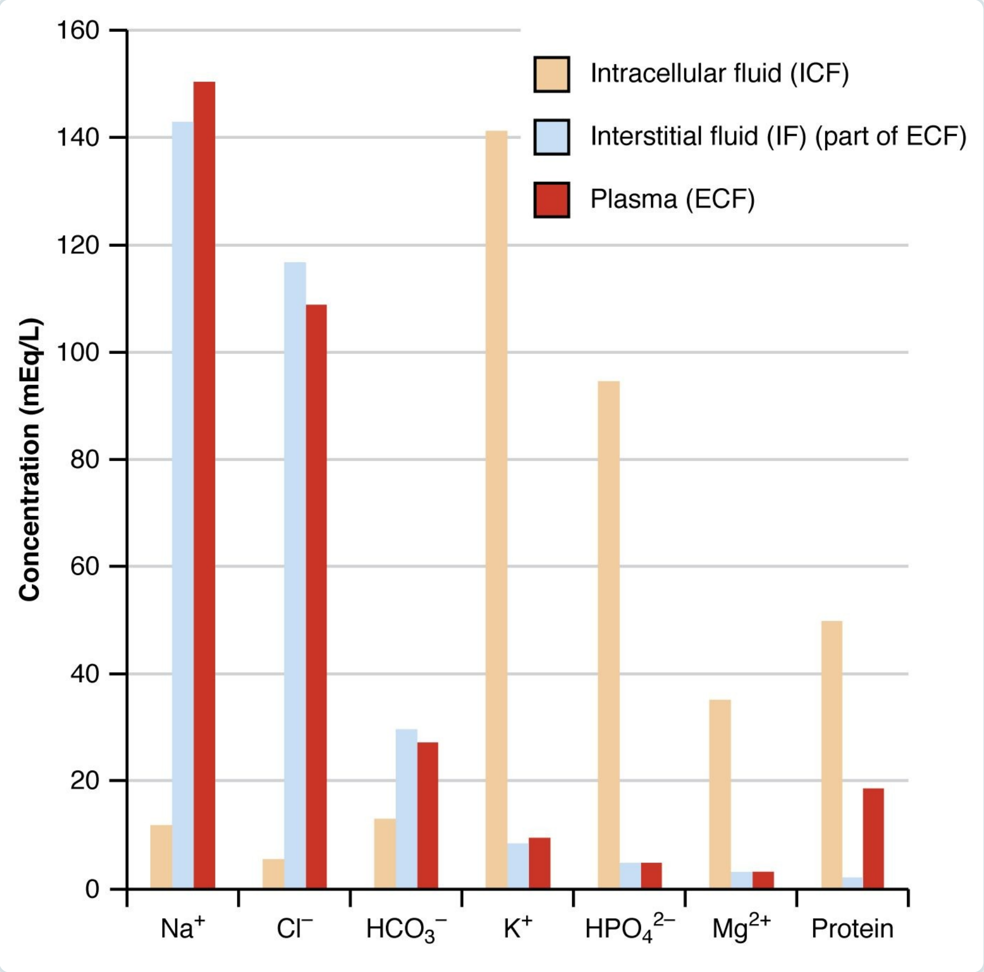
Electrolyte Balance

The electrolytes that drive nerve, muscle, and fluid balance.

SECTION 26.3

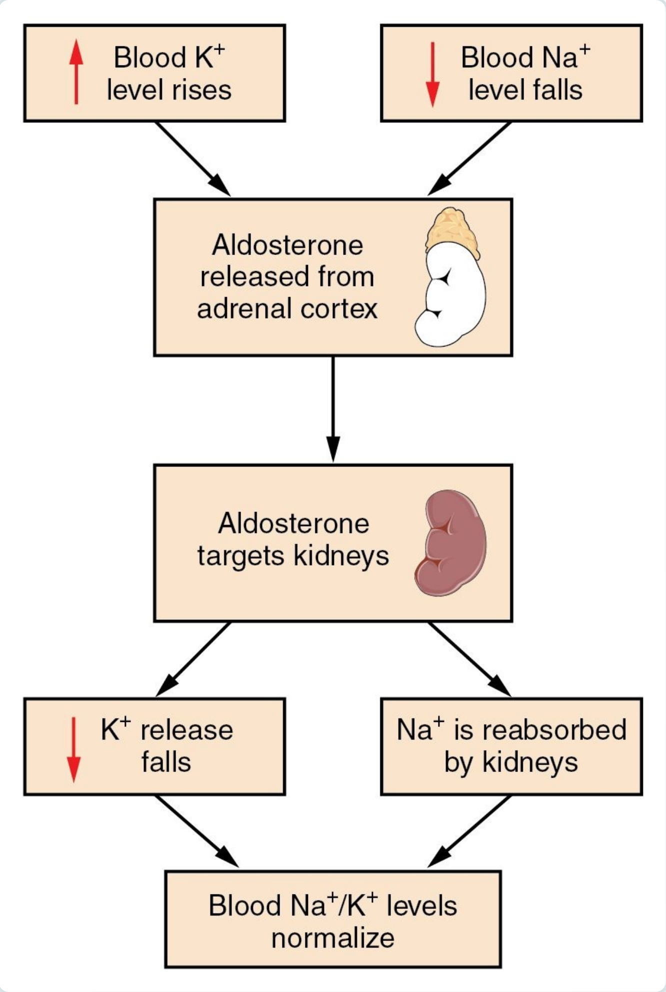
Electrolyte Composition

- Electrolytes are charged ions
- Concentrations differ between compartments
- Sodium dominates outside cells
- Potassium dominates inside cells



Electrolyte composition of body fluids

Sodium & Potassium Balance



Aldosterone and sodium-potassium balance

- Sodium and potassium are tightly controlled
- Aldosterone retains sodium
- And promotes potassium loss
- Imbalances affect the heart and nerves

26.4

SECTION

Acid-Base Balance

How the body keeps blood pH within a narrow safe range.

The pH Scale

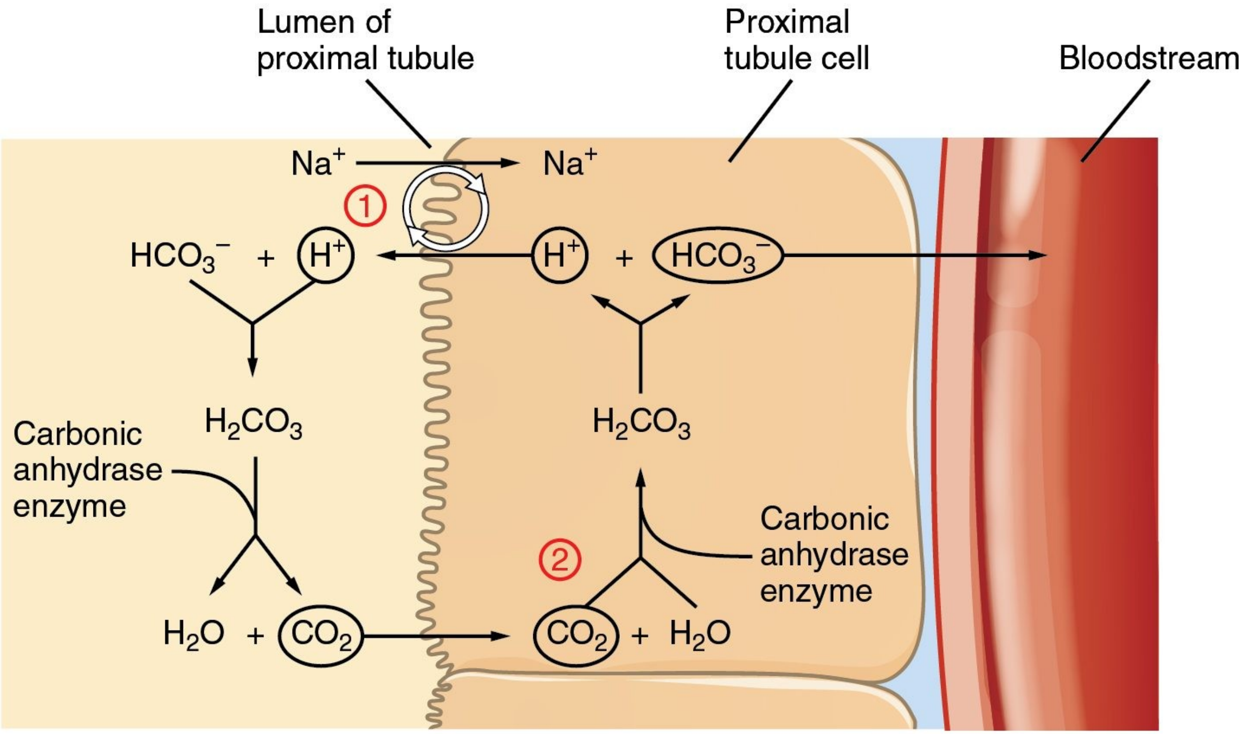
pH	Examples of solutions
0	Battery acid, strong hydrofluoric acid
1	Hydrochloric acid secreted by stomach lining
2	Lemon juice, gastric acid, vinegar
3	Grapefruit juice, orange juice, soda
4	Tomato juice, acid rain
5	Soft drinking water, black coffee
6	Urine, saliva
7	“Pure” water
8	Sea water
9	Baking soda
10	Great Salt Lake, milk of magnesia
11	Ammonia solution
12	Soapy water
13	Bleach, oven cleaner
14	Liquid drain cleaner

The pH scale

- pH measures acidity
- Blood is kept near 7.4
- Below 7.35 is acidosis
- Above 7.45 is alkalosis

SECTION 26.4

Regulating pH



The kidneys and acid-base balance

- Three systems defend blood pH
- Buffers act instantly
- The lungs adjust CO2
- The kidneys adjust bicarbonate and H+

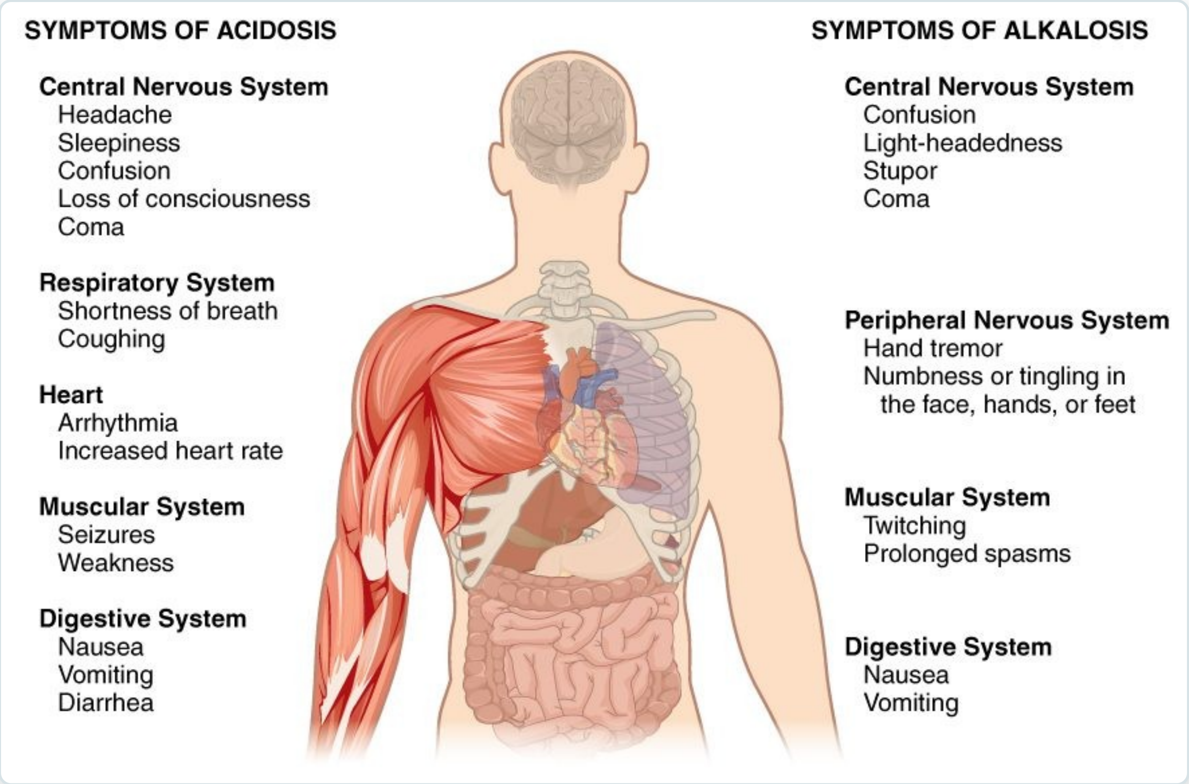
26.5

SECTION

Disorders of Acid-Base Balance

What happens when acid-base balance breaks down.

Acidosis & Alkalosis



Symptoms of acidosis and alkalosis

- Imbalance affects the whole body
- Acidosis: low pH (confusion, coma)
- Alkalosis: high pH (tingling, spasms)
- Can be respiratory or metabolic

Key Takeaways

- Body water is split between intracellular and extracellular compartments.
- Water balance matches intake to output, regulated mainly by the kidneys and ADH.
- Electrolytes differ between compartments and drive nerve, muscle, and fluid function.
- Blood pH is held near 7.4 by buffers, the lungs, and the kidneys.
- Acidosis and alkalosis disrupt the whole body and may be respiratory or metabolic.

CLINICAL CONNECTIONS

Why Fluid & Acid-Base Balance Matter in Practical Nursing

Dehydration

Fluid loss from vomiting, diarrhea, or fever needs prompt replacement.

Electrolyte Imbalance

Abnormal sodium or potassium can be life-threatening.

IV Fluids

Replace fluid and electrolytes when patients can't maintain balance.

Acid-Base Disorders

Respiratory and metabolic causes guide treatment.

Potassium & the Heart

High or low potassium causes dangerous arrhythmias.

Edema

Excess fluid in tissues signals a balance problem.